

RESTAURANT MANAGEMENT SYSTEM (RestMS)

PROJECT PROPOSAL DOCUMENT – ADVANCED VERSION

1. PROPOSAL SUMMARY

We propose to develop an **Advanced Restaurant Management System (RestMS)** with a professional website, full POS, third-party delivery integration (Zomato/Swiggy), AI-based menu recommendations, WhatsApp ordering, and comprehensive analytics. Ideal for **large standalone restaurants and small chains (5-15 outlets)** seeking complete digital automation and multi-outlet visibility.

Ideal For: Restaurant chains, large standalone restaurants, QSR franchises, cloud kitchen brands

2. OBJECTIVES

- Fully automate all restaurant and kitchen operations
- Integrate with Zomato, Swiggy, and other delivery platforms
- Deliver a seamless digital customer ordering experience
- Implement AI-powered menu recommendations and upselling
- Enable centralized multi-outlet management and analytics

3. SCOPE OF WORK

3.1 Website + Customer App (Advanced)

- Premium UI/UX restaurant website
- AI-powered menu recommendations
- Online ordering with live delivery tracking
- QR code table ordering (scan & order)
- Customer loyalty portal
- Google & Tripadvisor review integration

Cost: Rs. 45,000 - Rs. 70,000

3.2 Advanced POS & Billing System

- Full-featured POS terminal
- Multi-payment (Cash/Card/UPI/Wallet)
- GST-compliant billing with e-invoice
- Discount, coupon & happy hour management

- Void and refund management
- Shift-wise cashier reports

Cost: Rs. 50,000 - Rs. 75,000

3.3 Third-Party Delivery Integration

- Zomato & Swiggy order auto-sync
- Centralized order management across platforms
- Delivery status tracking
- Commission & payout tracking per platform

Cost: Rs. 40,000 - Rs. 60,000

3.4 Kitchen Display & Production System

- Multi-station kitchen display system
- Recipe & preparation guide integration
- Real-time order queue management
- Chef performance tracking

Cost: Rs. 30,000 - Rs. 45,000

3.5 Advanced Inventory & Cost Control

Recipe-based auto ingredient deduction

Supplier order management

- Wastage and spoilage tracking
- Food cost percentage analysis
- Purchase order generation

Cost: Rs. 40,000 - Rs. 60,000

3.6 Staff & HR Management

- Staff profiles with role management
- Attendance (manual / QR / Face recognition)
- Shift scheduling & leave management
- Tip distribution tracking
- Performance reports

Cost: Rs. 35,000 - Rs. 55,000

3.7 Admin Dashboard (Advanced Analytics)

- Multi-outlet real-time dashboard
- Item-wise sales & profitability reports
- Customer frequency & behavior analytics
- Staff performance insights
- Custom report export (PDF/Excel)

Cost: Rs. 35,000 - Rs. 50,000

4. ADDITIONAL FEATURES

4.1 WhatsApp Ordering & Notifications

- WhatsApp-based menu ordering
- Order status & bill on WhatsApp
- Promotional campaign via WhatsApp

Cost: Rs. 20,000 - Rs. 32,000

4.2 Customer CRM & Loyalty Program

- • Advanced loyaltypoints&tiers
 - Birthday / anniversary offers
 - Personalized offers based on order history

Cost: Rs. 20,000 - Rs. 30,000

4.3 Online Table Reservation System

- Advanced reservation withtime slots
- Guest preference management
- No-show and cancellation handling

Cost: Rs. 15,000 - Rs. 22,000

4.4 Role-Based Access & Security

- Multi-level roleswith granular permissions
- Secure JWT authentication

Cost: Rs. 12,000 - Rs. 18,000

4.5 Cloud Backup & Audit Logs

- • Dailyautomated backup
 - Complete activity and transaction logs

Cost: Rs. 12,000 - Rs. 18,000

5. TECHNOLOGY STACK (MERN + AI)

- **Frontend:** React.js
- **Backend:** Node.js (Express.js)
- **Database:** MongoDB
- **AI/ML:** Recommendation engine
- **Integrations:** Zomato/Swiggy API, WhatsApp Business API
- **Payment:** Razorpay / PayU
- **Hosting:** AWS Cloud

6. TOTAL PROJECT COSTING

Development Cost (One-time)	Rs. 4,50,000 - Rs. 6,50,000 (One-time)
Infrastructure	Hosting: Rs. 3,000 - Rs. 8,000/month
Infrastructure	Domain + SSL: Rs. 2,000/year
Infrastructure	WhatsApp Business API: Rs. 2,000 - Rs. 5,000/month
Maintenance Cost	Rs. 25,000 - Rs. 35,000/month/month

7. PROJECT TIMELINE

Phase	Duration
Planning	1 Week
Development	8-10 Weeks
Testing	1-2 Weeks
Deployment	1 Week
Total Duration	10-13 Weeks

8. LIMITATIONS

- No dedicated mobile app (can be added)
- No IoT integration with smart kitchen equipment
- No full franchise management module

9. FUTURE UPGRADES

- Mobile app (Customer + Staff)
- IoT smart kitchen integration
- Full franchise management
- AI chatbot for customer support

10. SUPPORT AND WARRANTY

- 1 month free support
- Bug fixing included
- Staff training provided

11. TERMS AND CONDITIONS

- 50% advance payment
- 50% after completion
- Additional features charged separately

12. CONCLUSION

This Advanced RestMS provides a **powerful, scalable, and highly automated platform** — integrating Zomato/Swiggy, WhatsApp ordering, AI recommendations, and multi-outlet analytics, making it perfect for restaurants ready for complete digital transformation.

13. CONTACT DETAILS

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